Medication Interaction Reference (BNF-Style Summary)

Purpose

This document provides a summary of key medication interactions relevant to the SignalLoop MedSafe rules engine. It covers the interactions most likely to be encountered in the demo scenarios and tested by judges.

Source: Based on BNF interaction guidelines (public severity ratings), Stockley's Drug Interactions summaries, and peer-reviewed literature.

Severity Classification

Severity	Definition	Action
Contraindicated	Must not be given together	Block prescribing
Major (Severe)	High risk of serious harm	Warn; require documented override
Moderate	Clinically significant but manageable	Warn; inform prescriber
Minor	Unlikely to cause problems for most patients	Inform only

Key Interactions

NSAID Interactions

NSAID +	Severity	Mechanism	Clinical Consequence
ACE-I/ARB + Diuretic	Major	Triple-whammy: removes all renal compensation	AKI (10-15% of elderly AKI admissions)

NSAID +	Severity	Mechanism	Clinical Consequence
ACE-I/ARB (without diuretic)	Moderate	Reduces renal perfusion	Reduced eGFR, hyperkalaemia risk
Warfarin	Major	Antiplatelet effect + GI mucosal damage	GI and other bleeding
Methotrexate	Major	Reduced renal clearance of MTX	Pancytopenia, hepatotoxicity
Lithium	Major	Reduced renal clearance of lithium	Lithium toxicity (levels rise 15-30%)
SSRIs	Moderate	Both impair platelet function	Increased GI bleeding risk (15x if combined)
CKD (eGFR <60)	Contraindicated	Prostaglandin-dependent renal perfusion	Accelerates CKD, precipitates AKI
Age ≥65 (chronic)	Moderate	Reduced renal reserve, GI vulnerability	GI bleeding, AKI, CV events (Beers)
Heart failure	Major	Fluid retention, prostaglandin inhibition	Acute decompensation

ACE-I/ARB Interactions

ACE-I/ARB +	Severity	Mechanism	Clinical Consequence
Potassium supplements	Major	ACE-I reduces K+ excretion	Hyperkalaemia (potentially fatal)
Spironolactone	Major	Both retain potassium	Hyperkalaemia (monitor closely)
Another ACE-I or ARB	Major	Dual RAAS blockade	AKI, hyperkalaemia (ONTARGET trial)
NSAIDs	Moderate-Major	See above	Reduced renal function

ACE-I/ARB +	Severity	Mechanism	Clinical Consequence
Trimethoprim	Moderate	Both increase potassium	Hyperkalaemia (especially in CKD)

Statin Interactions

Statin +	Severity	Mechanism	Clinical Consequence
Clarithromycin/Erythromycin	Major	CYP3A4 inhibition increases statin levels	Rhabdomyolysis risk
Grapefruit juice (large amounts)	Moderate	CYP3A4 inhibition	Increased statin levels
Fibrates (gemfibrozil)	Major	Pharmacokinetic + pharmacodynamic	Myopathy/rhabdomyolysis

Warfarin Interactions

Warfarin +	Severity	Mechanism	Clinical Consequence
Clarithromycin/Erythromycin	Major	CYP inhibition + altered gut flora	INR increase, bleeding
Trimethoprim	Major	Inhibits warfarin metabolism	INR rises 40-65%, bleeding
NSAIDs	Major	Antiplatelet + GI damage	Bleeding (see above)
Metronidazole	Major	CYP2C9 inhibition	INR increase

Cross-Reactivity (Allergies)

Allergy to	Cross-reaction risk with	Risk level
Penicillin	Cephalosporins	~1-2% (higher with 1st-gen cephalosporins)

Allergy to	Cross-reaction risk with	Risk level
Penicillin	Carbapenems	<1%
One NSAID	Other NSAIDs	High (COX-1 mediated, class effect)
Sulfonamide antibiotics	Sulfonamide diuretics	Low (different mechanism, but ask)

Renal Dosing Quick Reference

Drug	eGFR threshold	Action
NSAIDs	<60	Avoid (contraindicated)
Metformin	<45	Reduce to max 1g/day
Metformin	<30	Stop
Gabapentin	<50	Reduce max dose
Lithium	<60	Reduce dose, monitor levels frequently
Dapagliflozin	<25	Do not initiate (can continue if already on)